

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0918

Skiland 138 kV
November 16, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0918. The Interconnection Customer proposes a 114.4 MW natural gas combined cycle facility interconnecting at the Skiland 138 kV substation in the Pennsylvania area. The request was received on January 13, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 413 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,283,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CHEYENNE to KNOTT 138 KV Line	HANDSOME LAKE to PINEY CRK PROJECT 500 KV	114.0	216	5.1
ROCK TAVERN to CHEYENNE 138 KV line	HANDSOME LAKE SUBSTATION to PINEY CRK PROJECT 500 KV	106.0	236	5.1
LIGHTHOUSE HILL to STEWART MTN DAM SUBSTATION 345 KV	HILLTOP to PINEY CREEK PROJECT 500 KV	105.4	1,407	8.2
LIGHTHOUSE HILL to AES BEAVER VLY PARTNERS BEAVER VLY 345 KV	PINEY CRK PROJECT to HILLTOP 500 KV	103.7	1,576	8.6
ARMSTRONG to SOUTHEAST VICKSBURG 500 KV Line	HILLTOP to PINEY CREK PROJECT 500 KV	103.2	2,185	7.4
MONTOUR STATION SUBSTATION to ACADIA 500 KV	CORONET to WESTWOOD GENERATION LLC 500 KV L	101.0	2,796	8.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHEYENNE to KNOTT 138 KV	Rebuild line with higher-capacity conductor	\$16,175,000	21% (\$3,391,000)
CHEYENNE to ROCK TAVERN 138 KV	Reconductor line	\$506,000	21% (\$106,000)
LIGHTHOUSE HILL to STEWART MTN DAM SUBSTATION 345 KV	Reconductor line	\$12,831,000	2% (\$312,000)
LIGHTHOUSE HILL to AES BEAVER VALLEY PARTNERS BEAVER VALLEY 345 KV	Reconductor line	\$4,647,000	2% (\$108,000)
ARMSTRONG to SOUTHEAST VICKSBURG 500 KV Line	Reconductor line	\$30,080,000	2% (\$568,000)
MONTOUR STATION to ACADIA 500 KV	Reconductor line	\$35,916,000	2% (\$798,000)
	Total allocated to project		\$5,283,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.